# Checkpoint 1 Report

CS215 — 21 Oct 2026

## Section 1: Meta Information

| | |
|---|---|
| **Project title** | What Explains the Rise of Indian Chess? |
| **Group number** | "25" |
| **Mentor** | Mayank Jain |
| **Members** | Rane (25B0975), Dhruv Parsewar (25B0977), Karmanya Gupta (25B1069), Shiven (25B1089) |
| **GitHub repo** | `https://github.com/gitbromamide/Team_25.git` |

## Section 2: Progress So Far

### 2.1 Dataset and descriptive statistics

We use the FIDE monthly rating lists (XML) in three formats, *standard*, *rapid* and *blitz*, from `ratings.fide.com/download_lists.phtml`. The lists are public and need no login. FIDE's archive starts in Feb 2015 (our proposal said 2001; this is corrected). We use the January and July lists for 2015–2026 plus the current Oct 2026 list, giving 24 lists per format. Each row has 7 source fields (FIDE ID, federation, sex, title, rating, birth year, flag) plus our `snapshot` and `format` columns.

| Format | Lists | Players per list (min–max) | Total rows |
|---|---|---|---|
| Standard | 24 | 214,727 – 569,654 | 9.12M |
| Rapid | 24 | 51,100–479,701 | 5.47M |
| Blitz | 24 | 41,785–317,994 | 3.83M |

We then aggregate to one row per format, list date and federation . It holds 12 measures: rated players; players above 2000/2300/2500/2700; IM+GM and GM counts; and mean, top-10 and top-100 ratings.

### 2.2 Data cleaning completed

(i) Streaming XML parse of each list (about 0.5M players each) into Parquet, one file per format and date.
(ii) Format and month read from file names; the rating field is taken from the format-specific tag.
(iii) Numeric coercion; unrated players (no rating) excluded from all counts.
(iv) Inactive players flagged via the `flag` field (provisional; see blockers).
(v) Lists that are not XML are skipped and logged.
(vi) Raw files are not committed; scripts rebuild them from FIDE.

### 2.3 Division and revision of responsibilities

| Member | Current responsibility |
|---|---|
| Rane | Download/parsing pipeline and data audit |
| Dhruv | Data visualization using matplotlib, numpy, pandas |
| Karmanya | Repository management, spreadsheet management |
| Shiven | Analysing post-event jumps (scipy.stats) |

### 2.4 Fine-grained problem statement

Using the standard, rapid and blitz lists (Feb 2015–Oct 2026), we ask four questions.

1. Is the growth in India's players rated 2300+ and 2500+ explained by a larger rated pool or by a higher share of strong players?
2. Does growth accelerate after March 2020 (online chess), the 2022 Olympiad or Gukesh's 2024 title, relative to a pre-existing trend?
3. Is India's share of the world's strong players rising, and is this unusual compared with China, the USA, Russia and others?
4. Do the three formats tell the same story, given that rapid and blitz may have reacted differently to online chess?

**First look (standard, Jan 2016 to Jan 2026).**

| India, standard list | Jan 2016 | Jan 2026 | Change |
|---|---|---|---|
| Rated players (pool) | 20,123 | 51,859 | 2.6× |
| Players rated 2300+ | 141 | 281 | 2.0× |
| Share of pool rated 2300+ | 0.70% | 0.54% | fell |

The 2300+ share of the pool fell, which favours a participation explanation over a strength explanation. Pool growth slowed in 2020–2022 and picked up from 2023.

### 2.5 Updated tracking spreadsheet

`https://docs.google.com/spreadsheets/d/1tvUrJxv8OnBDu2KosB9XRdEo8XhPkGPStZWP895Ao9U/edit?usp=sharing`

- Meta-information row added.
- Edit access shared with our mentor.

### 2.6 Blockers / challenges

1. **Active-player definition.** Active counts for India dip sharply in Jan 2022 (12,451 to 5,444) and recover. We suspect a change in how `flag` is used or in FIDE's inactivity rule but have not confirmed it. Should we use all rated players or only active ones?
2. **Usage terms.** FIDE's terms for the lists need confirming before the final report.
3. **Statistical testing.** We need guidance on testing whether post-event changes exceed normal year-to-year variation, with only 2 snapshots a year.

### 2.7 Plan for Checkpoint 2 (4 Nov)

- Build pre/post-event trend comparisons (2020, 2022, 2024) with uncertainty estimates.
- Add benchmark federations (China, USA, Russia) and compare India's share of strong players.
- Produce final charts, the 3–5 minute video, references and the reflection note.